

Brooks A. Benard, Ph.D.

Computational Biologist

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SUMMARY

As a computational biologist with extensive experimental experience, I leverage atlas-scale data into experimentally actionable insights across domains such as cancer, muscular dystrophy, and Alzheimer's disease. My research spans bulk, single-cell, and spatial omics, with expertise in clinical genomics and transcriptomics, clonal evolution, drug-sensitivity modeling, and target nomination. I build tools, databases, and methods (including PRECOG, PhenoMapR, and Shiny applications) that convert large-scale analyses into results that experimental biologists can act on. This dual background places me at the intersection of experimental and computational biology, where I help shape experiments and translate findings into clinically actionable results.

EXPERIENCE

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Jul 2023 – Present

Stanford University School of Medicine, Gentles Lab, Department of Pathology · Stanford, CA

- Curated and shipped **PRECOG 2.0**, a clinical-outcomes and gene-expression database covering **335 datasets, >46,000 patients, and ~55 cancer malignancies** (adult, pediatric, immunotherapy). It is the largest database of gene expression and cell-type abundance associations with clinical outcomes across cancer to date (Benard et al. NAR 2025).
- Developed **PhenoMapR**, a method and R package for semi-supervised mapping of sample phenotype associations onto single-cell, spatial, and bulk transcriptomic data; validated across disease areas including **cancer, muscular dystrophy, and Alzheimer's**. *Manuscript in preparation*.
- Applying PhenoMapR **at scale** across single-cell and spatial **pan-cancer atlases** (TISCH2, scPCA, HTAN; >10M cells) to prioritize cellular subpopulations for clinically-actionable experimental validation (Stanford Cancer Institute AI Symposium, 2025).
- Design analyses and prioritize hypotheses alongside experimental and clinical collaborators; communicate findings to wet-lab biologists and quantitative/ML collaborators so results immediately inform the experiments and research directions.

Ph.D. Candidate, Cancer Biology

Sep 2017 – Jun 2023

Stanford University, Majeti Lab · Stanford, CA

- Led end-to-end research on how **clonal architecture in AML** relates to prognosis and drug response, spanning question framing through analysis, interpretation, and release of tools/data (Benard et al. Nat Commun 2021).
- Assembled and analyzed one of the largest molecularly annotated AML cohorts (**~2,800 patients**), modeling variant allele frequency, mutation order, and evolutionary patterns against clinical outcomes and drug sensitivities on large genomic datasets.
- Built **Meta AML Explorer**, an interactive Shiny application for AML mutation-outcome exploration across ~4,000 patients.
- Worked alongside experimentalists to design follow-up and provide computational validation for wet-lab projects (IL-6/bone-marrow failure, mutation-order evolution, drug-sensitivity and target mining), scoping ambiguous questions into actionable results.
- Authored a review on data mining for mutation-specific therapeutic targets in AML (Benard et al. Leukemia 2019).

Research Associate I (Keats Lab)

May 2016 – Sep 2017

Translational Genomics Research Institute (TGen) · Phoenix, AZ | Undergraduate Research / Helios Scholar, 2014–2016

- Contributed genomic analysis for the **MMRF CoMMpass** study, the largest longitudinally sequenced and clinically annotated study in multiple myeloma to date; co-authored manuscript in *Nature Genetics* (2024).
- Wrote software still used to automate clinical-outcomes analyses from patient classification features; identified high- and low-risk genetic subsets (FGFR3 as an adverse prognostic factor; ASH, 2017).
- Experimental biology:** assessed multiplexed PCR assay sensitivity for cell-line purity; predicted and **experimentally validated** chemotherapy-repurposing susceptibility via drug-dosing in myeloma cell lines; benchmarked gene-fusion callers.

Undergraduate Research Assistant (Haydel Lab)

Aug 2013 – Dec 2014

ASU Biodesign Institute, Center for Infectious Diseases and Vaccinology · Tempe, AZ

- Experimentally characterized** DNA-binding affinities for proteins in the *M. tuberculosis* PrrAB two-component system; early studies that helped lay groundwork for later published work on druggable interactions.

EDUCATION

Ph.D., Cancer Biology, Stanford University | Advisor: Ravindra Majeti, M.D., Ph.D.

Dissertation: *Integrating features of clonal evolution in modeling disease risk and drug sensitivity in AML*

B.S., Molecular Biosciences & Biotechnology, Arizona State University

SELECTED PUBLICATIONS

1. **Benard BA**, Lalgudi CK, Ilerten I, Wang RH, Gentles AJ. PRECOG update: an augmented resource of clinical outcome associations with gene expression for adult, pediatric, and immunotherapy cohorts. *Nucleic Acids Research*. 2025.
2. **Benard BA**, Leak LB, Azizi A, Thomas D, Gentles AJ, Majeti R. Clonal architecture predicts clinical outcomes and drug sensitivity in acute myeloid leukemia. *Nature Communications*. 2021.
3. Skerget S, ... **Benard B**, ... Keats JJ, et al. Comprehensive molecular profiling of multiple myeloma identifies refined copy number and expression subtypes. *Nature Genetics*. 2024.
4. Schwede M, ... **Benard B**, Gentles AJ, ... Majeti R. Mutation order in acute myeloid leukemia identifies uncommon patterns of evolution and illuminates phenotypic heterogeneity. *Leukemia*. 2024.
5. Zhang TY, Dutta R, **Benard B**, Zhao F, Yin R, Majeti R. IL-6 blockade reverses bone marrow failure induced by human acute myeloid leukemia. *Science Translational Medicine*. 2020.
6. **Benard B**, Gentles AJ, Köhnke T, Majeti R, Thomas D. Data mining for mutation-specific targets in acute myeloid leukemia. *Leukemia*. 2019.
7. Boutelle AM, ... **Benard BA**, ... Attardi LD. Integrative multiomic approaches reveal ZMAT3 and p21 as conserved hubs in the p53 tumor suppression network. *Cell Death & Differentiation*. 2025.
8. Karigane D, ... **Benard BA**, ... Majeti R. Intra-leukemic interferon signaling suppresses expansion and mediates chemoresistance in human AML. *Blood Cancer Discovery*. 2025.

TOOLS & RESOURCES

PRECOG	Prognostic/predictive gene & cell-type associations across cancers and treatment modalities · precog.stanford.edu
PhenoMapR	Method, R package, and Shiny app for mapping sample phenotype signals onto single-cell, spatial, and bulk transcriptomes · brooksbenard.github.io/PhenoMapR
Meta AML Explorer	Shiny app for interactive exploration of AML mutational associations with outcomes and drug sensitivity · brooks-benard.shinyapps.io/MetaAML_Shiny
DepMap Shiny	Interactive exploration of mutational and cancer dependencies across the Broad's Dependency Map CRISPR screening database · brooks-benard.shinyapps.io/DepMap_Shiny

TECHNICAL SKILLS

Computational biology: Bulk RNA-seq & clinical outcomes; single-cell & spatial transcriptomics; clonal evolution / VAF modeling; comparative genomics; immune deconvolution; survival & response modeling; drug-sensitivity / high-throughput screen analysis; hypothesis generation from literature and curated knowledge bases

Software & computing: R (primary; tidyverse, Bioconductor, Shiny), Python; analysis pipelines & cohort databases; large datasets on Linux/HPC; interactive tools for biologists; reproducible analysis packages; agentic systems and workflows

Experimental biology: Cell culture; shRNA knockdowns; western blots; cloning; protein-DNA binding assays; cell-line genomic QC & multiplex PCR; drug-dosing / chemosensitivity; mouse work (intrafemoral injections, embryonic dissections, etc.); confocal microscopy; flow cytometry and FACS; designing experiments and interpreting results alongside experimentalists

SELECTED PRESENTATIONS & HONORS

Presentations: Pan-cancer cell states associated with prognostic risk and therapy response at single-cell and spatial resolution, presented at the Stanford Cancer Institute AI Symposium (2025), Stanford Cancer Institute Research Retreat (2025), Stanford Center for Cancer Systems Biology NCI Symposium (2024 & 2025), and Stanford Pathology Department Research Retreat (2024)

Honors: Blavatnik Fellow (Stanford Medicine, 2021); ASH Abstract Achievement Award (2020); Excellence in Service to Graduate Students (Stanford Biosciences, 2020); NSF GRFP Honorable Mention (2018); ASH Abstract Achievement Award (2017); ASU *Summa Cum Laude* (2016); Student of the Year, Molecular Biosciences & Biotechnology (2016)

CONSULTING

Balsam Bio · Inograft Biotherapeutics · Ideaflo: computational analyses, tool building, and biological input that directly informed target selection and experimental direction for company biologists.